

NATIONAL UNIVERSITY

B.Sc. (Hons.) in Computer Science and Engineering
First Year, Second Semester Final Examination, 2021
Course Code: 510222
Course Title: Digital Systems Lab

Time: 3 Hours

Full Marks: 40

Answer any one (01) of the following questions:

1. Verify the truth tables of logic gates (OR, AND, NOT gate).
2. Verify the operation of NAND gate as universal gate.
3. Verify the function of XOR gate using basic gates.
4. Design and implement a half adder circuit and verify its operation.
5. Design and implement a half subtractor circuit and verify its operation.
6. Design and implement a decoder circuit.
7. Simplify and implement the following logic expression:
$$Y = \bar{A}BC + ABC + ABC$$
8. Design and implement a Binary-to-gray code converter.
9. Design and implement a 4:1 Multiplexer.
10. Design and implement an S-R flip flop.
11. Design and implement a 2-bit asynchronous binary counter.
12. Design and implement a 4-bit D/A converter using binary weighted register method.

Marks Distribution:

1. Circuit Design	: 10
2. Circuit Implementation	: 10
3. Results Analysis	: 05
4. Viva	: 15
Total Marks	: 40

Handwritten signatures and initials:
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